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Characterization of the artificial intelligence labor market in Chile: An approach based on unstructured data and the ESCO taxonomy

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ABSTRACT

This study characterizes the labor demand related to artificial intelligence (AI) in Chile to identify the skills and requirements sought by employers. Using unstructured data from online job postings collected from major employment platforms, AI-related vacancies are identified and analyzed through a skills-based approach aligned with the European Skills, Competences, Qualifications, and Occupations (ESCO) taxonomy. The results show that demand is primarily oriented towards technical skills, while soft skills play a complementary and transversal role, with teamwork being the most recurrent. AI-related job demand is concentrated in high-skilled occupations and specialized professional profiles, while patterns observed in the data are consistent with skill-biased and routine-biased technological change hypotheses. The identified skill structure highlights the relevance of advanced digital competencies for emerging AI occupations and provides a basis for updating training programs. Overall, the findings offer empirical evidence on the early diffusion of AI-related jobs demand in Chile and contribute to the discussion on skills development and labor market adaptation in emerging economies.

1. Introduction

In recent years, artificial intelligence (AI) has emerged as a powerful alternative to improve efficiency, productivity and business optimization in many productive activities and services (Liu et al., 2022; Nath et al., 2024). The existing conceptualizations of AI include definitions as simple as it being computer systems with human-like intelligence (Wierenga, 2010), or simply the replication of human intelligence by a system or machine (Nath et al., 2024). Furthermore, such systems are said to develop abilities 'to correctly interpret external data, learn from that data, and use those learnings to achieve specific goals and tasks through flexible adaptation' (Haenlein & Kaplan, 2019).

As a result, AI has been increasingly incorporated by companies. According to the annual report on the state of AI worldwide, 20 % of companies reported using AI in at least one of their functions in 2017, while, by 2022, this percentage had risen to 50 %. In other words, the global adoption of AI multiplied by 2.5 in a 5-year period (Chui et al., 2022).

As AI advances, concerns have emerged regarding the effects of its implementation on the labor market. While automation may reduce labor demand in tasks that are directly automated, productivity gains may generate a compensatory effect, leading to increased labor demand in non-automated tasks (Acemoglu & Restrepo, 2019a, 2019b; Yang, 2022).

In particular, the skill-biased technological change (SBTC) hypothesis suggests that a change in production technology favors skilled over unskilled labor (Violante, 2008). The expansion of AI can increase labor market inequalities by increasing the demand for skills linked to highly skilled jobs, while it can have a negative impact on other types of less skilled jobs (Holm & Lorenz, 2022). Instead, Wang et al (2021) illustrate that SBTC may increase the demand for high-skilled workers and also for low-skilled workers, when they are complementary (Wang et al., 2021). It has been also posited that the substitution of low-skilled workers performing routine activities using AI processes, such as robotics and automation, leads to an increase in the proportion of technology-intensive industries, which contributes to improving the

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skill structure (Hu et al., 2024) (i.e. that workers with less skilled jobs and competencies may increase their skillsets in order to achieve higher skills and competencies). On the other hand, the routine biased technological change (RBTC) hypothesis affirms that information technology (IT) tends to replace workers focused on routine activities (Hang et al., 2024; Martins-Neto et al., 2024). In other words, '*there is a shift in relative demand away from occupations that are more routine*' (Goos et al., 2014).

Amid the trade-offs associated with the benefits and risks of AI, a clear reality has emerged: AI is becoming increasingly embedded across multiple dimensions of modern life. This shift reframes the debate about whether AI should be adopted to how countries can effectively prepare to address the challenges it poses. In this context, understanding the dynamics that characterize AI-related jobs is essential for responding to emerging training needs and for supporting educational institutions, firms, and public agencies responsible for workforce development and employment regulation.

2. Literature review

Analyzing the impact of AI on automation and the labor market requires an understanding of the scope and characteristics of its main subfields, summarized in Table 1.

In general, these subfields allow us to infer the skills potentially demanded by employers in AI, although their specification may vary depending on the technological and geographical context and the taxonomy of specific skills.

The characterization of the labor market in relation to occupational profiles linked to technological change, especially AI, has been addressed in the literature in a small group of countries in recent years. Studies focusing on the United States of America (USA) stand out, such as the case of Gardiner et al. (2018) who analyzed 1,216 online job advertisements related to 'big data'. This study emphasized that both analytical information systems and soft skills are highly valued, while emerging technology skills also present significant added value (Gardiner et al., 2018). Or the one launched by Verma et al. (2019) examined a sample of online job postings in the same country. The profiles analyzed were those related to professions such as business analyst (BA), business intelligence analyst (BIA), data analyst (DA) and data scientist (DS). This allowed a list of relevant skills for the positions studied to be obtained, where technical skills such as statistics and programming skills were the most demanded (Verma et al., 2019; Verma et al., 2019). Also, Verma et al. (2022) used data from the online job portal Indeed.com to collect job ads related to AI and ML, to explore the skills needed for these positions. The data were obtained by web scraping (using Python for titles that included AI and ML). In this study, it finds that machine learning roles emphasize deeper technical competencies while AI roles value more general and communicative skills (Verma et al., 2022).

Some relevant gaps in the literature, particularly regarding wage premiums and demand trends associated with AI skills, are addressed by Alekseeva et al. (2021). Using the Burning Glass Technologies online job database from 2010–2019, they analyzed the demand for AI skills across US companies and occupations over time and across companies, occupations, and industries. They revealed a significant increase in the number of jobs requiring AI skills and demonstrated that these skills have a positive effect on wage premiums. They also identified that jobs requiring customer service skills show evidence of being replaced by AI (Alekseeva et al., 2021).

While Alekseeva et al. (2021) document the concentration of AI-related demand in high-skilled occupations, they do not explicitly address employment polarization or regional inequalities. Furthermore, the use of a proprietary skills classification framework limits comparability across contexts. This introduces restrictions like those observed in consensus-based approaches (Gardiner et al., 2018). These limitations are addressed by Colombo et al. (2019), who analyze online job postings requiring AI skills in Italy to examine the relationship between skills demand and automation risk. The authors apply a competency classification system aligned with the European Skills, Competences, and Occupations (ESCO) framework, which improves comparability across contexts and enables large-scale empirical analysis. In addition to documenting the growing demand for advanced digital and social skills, the study explicitly addresses labor market polarization by showing that occupations requiring higher cognitive and social skills face a lower risk of automation. It also reveals territorial disparities regarding the demand for jobs requiring AI skills (Colombo et al., 2019).

Evidence on the demand for AI-related skills in Latin America is scarce. One of the few empirical contributions in the region is the study of Mamani (2022), which analyzes online job postings for Information Technology (IT) professionals in Peru using web scraping techniques and an unsupervised machine learning approach. Based on 8,640 job offers, the study identifies occupational profiles and technical and soft skills. To do this, it uses natural language processing and clustering methods. While this approach represents a valuable methodological contribution for data-scarce contexts, the analysis focuses on general IT occupations rather than specific AI roles. The same limitations remain in

Table 1
Main AI subfields.

AI subfields	Scope	Citation
Neural networks	They are used for brain modeling, time series prediction and classification, and can simulate human cognitive abilities for problem solving.	(Ossowska et al., 2022)
Evolutionary computation	It is used to solve problems that have too many variables for traditional algorithms. It covers gene expression programming algorithm, symbolic regression, machine learning (ML) models, among many others.	(Muhammad et al., 2021; Oduguwa et al., 2005)
Artificial vision	It encompasses object recognition and image understanding, which aims to mimic human vision. It includes fields such as computer vision and face recognition.	(Chang et al., 2024; Poddar et al., 2024)
Robotics	It includes intelligent control and autonomous exploration, being '*able to perceive, reason, learn, make decisions and execute tasks without direct human control, capable of mimicking and executing various tasks of human intelligence*'.	(Li et al., 2024; McKendrick et al., 2021)
Expert systems	They cover everything from decision support systems to image processing, classifications, etc.	(Qiao, 2024)
Speech processing	It deals with speech recognition and production, natural language processing, used in machine translation.	(Escobar-Grisales et al., 2023; Jung et al., 2020)
Automated planning	It seeks to create computer programs that can generate plans to achieve a particular objective, including the programming of tasks and games.	(Gan et al., 2023; Geist, 2017)
Machine Learning (ML)	Algorithms are used to give computers the ability to identify patterns in massive data and make predictions (predictive analytics), allowing computers to learn autonomously and perform actions without the need to be programmed.	(Mamani, 2022; Moura et al., 2024; Suresh Kumar et al., 2024)

Source: Own elaboration, 2025.

terms of the use of ad hoc classifications rather than standardized taxonomies, and it does not explicitly address broader labor market dynamics such as employment polarization, regional inequalities, or sectoral transformation.

In summary, there are significant gaps in the literature on the demand for employment related to AI skills in emerging economies and their territorial implications. This gap, together with the limited application of standardized classification frameworks such as ESCO or comparable approaches (Colombo et al., 2019), highlights the need to generate empirical evidence that systematically addresses these dimensions and contributes to the broader debate on employment polarization and regional inequalities.

Anticipating labor gaps is a key necessity in Latin American countries to improve the employability and quality of life of the population (Gontero & Albornoz, 2019). In this context, Matamala et al. (2024) analyze the role of Chilean universities in the development of digital skills, highlighting persistent challenges associated with gender gaps and family educational levels (Matamala et al., 2024). In this context, the objective of the present research is to provide a characterization of AI related employment in Chile, based on unstructured open data from the online job postings of the most widely used web portals in the country, and the use of the ESCO taxonomy. The study also contributes to the literature on skill-biased and routine-biased technological change in three ways. First, rather than testing these frameworks causally, it provides descriptive empirical evidence on how patterns consistent with SBTC and RBTC emerge in the early diffusion of AI-related labor demand within an emerging economy. Second, it extends these frameworks to a territorially heterogeneous context, highlighting how AI-related job demand is unevenly distributed across regions and sectors. Third, by focusing on online job postings, the study offers a timely perspective on employers' skill requirements, complementing existing research based on employment data. To fulfill the objective of this research, the rest of the article is structured in a methodological section (Section 3), followed by the results (Section 4). A discussion was generated from the results obtained, that seeks to contribute to the discussion on the SBTC and RBTC hypothesis (Section 5). Finally, the paper concludes (Section 6) and includes the limitations of the study.

3. Methodology

A data collection and analysis methodology was established to achieve the objective, and which consisted of four phases: Data source selection; Collection of unstructured open data from the selected sources; Data construction and processing; and Visualization and analysis (see Fig. 1).

It is important to note that this study analyzes AI-related labor demand as expressed in online job postings, which are recruitment signals issued by companies in response to perceived staffing needs. While job postings do not represent actual employment outcomes, they are consistent with the goal of anticipating emerging skills needs, particularly in the early stages of technology adoption, when changes in labor demand tend to materialize first through employers' hiring intentions rather than through observable employment statistics. Therefore, our results identify patterns of skill demand that could be consistent with the hypotheses described above, if such hires were to materialize.

The following is a detailed description of how each of these phases was carried out:

Phase 1: Selection of the data source.

Four online job portals were selected (i.e. CompuTrabajo, LinkedIn, Trabajando.cl and ChileTrabajos.cl). These sites were chosen according to three hierarchical criteria (in terms of representativeness and amount of available information) (Gontero & Menéndez, 2021): i) priority for the most visited websites in Chile in the category of job portals, according to Similarweb metrics (2023); ii) Websites where the most vacancies related to AI appeared during the systematic review process (exploration); and iii) data that contain a significant amount of relevant information in terms of the identified variables. Although LinkedIn was not among the most visited job portals in Chile, it was chosen for its inclusion due to its satisfactory fulfilment of the rest of criteria. No metasearch engines were included to avoid duplicity in the advertisements.

Overall, a total of 1,100 online job vacancies were collected during the three months of monitoring. Out of these, 45.45 % were obtained on LinkedIn, followed by CompuTrabajo (28.18 %), ChileTrabajos (15.73) and, finally, Trabajando (10.64).

Phase 2: Collection of unstructured open data from the selected sources

The data collection was carried out systematically every day in 2023 from August 5 to November 5, using the Web scraping technique in the same way as other studies in this field (Gontero & Menéndez, 2021; Mamani, 2022). An in-depth review of the literature and the analysis of the exploratory results was launched with the purpose of optimizing the search process for job demands related to AI, as a result, the following keywords were established as search criteria: 'Artificial intelligence' 'Prompt or prompt engineer' 'Neural networks' 'Machine Learning' 'Deep learning' and 'IA or AI' (in two languages, English and Spanish). As mentioned above, this made it possible to collect information on 1, 100 vacancies. These numbers are considered suitable for content analysis methodologies (Gardiner et al., 2018; Verma et al., 2019, 2022).

Phase 3: Database processing and construction

It is important to highlight that the advertisements have no standard structure for the different websites, which makes the processing and construction of the database more complex. In this sense and based on the data collected in the previous step, four activities were carried out: cleaning the data, identification of variables; construction of the database, and identification of skills. The following is a description of these activities:

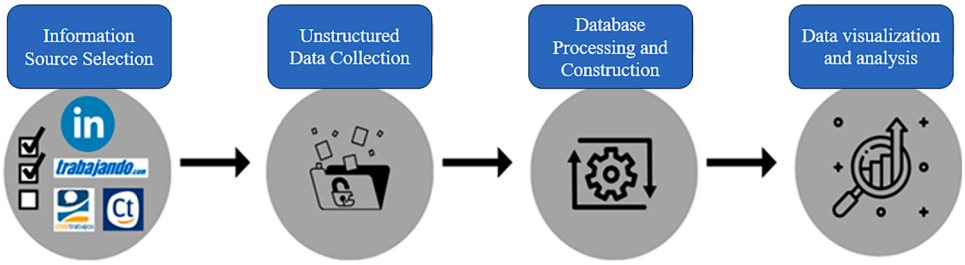

Fig. 1. General methodological outline. **Source:** Own elaboration, adapted from Gontero and Menéndez (2021).

a. *Cleaning*: first of all, a cleansing task was performed in order to eliminate duplicities (i.e. publications on the same date, by the same company on different portals, etc.).
b. *Identification of standardized variables*: with the information collected and cleaned, we proceeded to structure the data in standardized variables present in the descriptions and headings of the online job vacancies. These variables are the following: job title, salary, workday (full, part or on-call), modality (on-site, remote or hybrid), name of employer (company), industry, firm size, geographic location, skills required¸ level of education, type of degree required and years of experience. The latter are used to identify whether the demand for employment is for highly skilled, moderately skilled, or low-skilled positions.
c. *Construction of the database:* a relational database approach was chosen, according to the requirements inherent to the information being managed. The database was created using Microsoft Access software, which offered the ability to establish various levels of cardinality in the relationships between entities.
d. *Identification and Classification of Skills and occupations*: the ESCO taxonomy is used (European Commission, 2024). In this process, the skills identified in the online vacancies were categorized by using keywords corresponding to this taxonomy, accompanied by their respective codes. The ESCO classification divides the skills and competencies into four types: K: Knowledge; L: Linguistic competencies and knowledge of languages; S: Skills; and T: Transversal competencies. In turn, the skills classified under "K'' belong to the category of 'hard-skills', while the skills contained in "L'' and "T'' are categorized as 'soft-skills'. On the other hand, skills classified as 'S' can belong to any of the categories (Colombo et al., 2019).

The use of the ESCO taxonomy addresses the lack of a standardized skills classification framework for Latin America, being a recurring limitation in the literature. The problem of ad hoc or context-specific classifications is that they hinder consistent comparisons between countries. It builds upon the International Labour Organization's International Standard Classification of Occupations (ISCO) and incorporates a structured skills and competences pillar within a multilingual framework developed by the European Commission (European Commission, 2024), enabling transparent and comparable analyses across different contexts. While it was initially developed for the European Union and being potentially less sensitive to locally adapted terminology in Chile, we consider that these limitations could primarily affect the granularity of skills, rather than the identification of broader demand patterns.

Phase 4: Data visualization and analysis

In this phase of the study, the data were exported from Microsoft Access to IBM SPSS. This software facilitated the visualization of the data (which had previously been cleaned and structured; see Phase 3-b). The variables selected for the descriptive representation in the graphs integrate all the aspects related to the organization that presents the labor demand and the profile of the worker required according to the advertisement fields. Descriptive frequency analyses were conducted for qualitative variables, while quantitative variables were summarized using basic statistical measures. SQL queries were used solely to aggregate and filter data according to analytical needs, without affecting the underlying structure of the dataset. Given the poor standardization of titles, Word Cloud for Google Chrome tool was used to make a word cloud of the post titles to achieve their graphical grouping. Due to extensive missing information in salary fields (only 64 vacancies indicated it), compensation-related analyses were excluded.

4. Results

4.1. General characterization of the job vacancies

Regarding the companies that advertise AI related jobs, Fig. 3 shows that the commune of Santiago (with 40.7 % of the publications) emerges as the epicenter of AI-related job vacancies in the dataset. A high concentration of demand is evident in the Metropolitan region of the country (Santiago, Las Condes, Providencia, San Bernardo, Lampa, and Huechuraba) with more than 56.8 %, which could be related to the fact that the capital city tends to attract a higher density of companies and job opportunities, as the center of economic and business activity in Chile.

It is striking that the location "Chile'' occupies 22.4 % of the observations. This result suggests that many of the vacancies are offered in a format that allows working remotely or that the exact location of the job seeker is not a critical or excluding factor for filling these positions. On the other hand, only three municipalities outside the national capital (Concepción (7.82 %), Viña del Mar (6.09 %) and Valparaíso (2.36 %)) show a demand of more than 1 %.

However, it was determined that 50.7 % of companies seeking employees related to AI establish the requirement to work remotely. The other modalities, such as hybrid and face-to-face are required by 25.8 % and 22.7 % of the vacancies respectively. Although these results can be attributed to the characteristics of this type of jobs, it could also be related to the changes caused by the impact of Covid-19 (Lopes & Alves, 2024).

Similarly, 95.82 % of the demand corresponds to full-time work. Only the part-time and shift work modalities are sought at 2.27 % and 1.91 % respectively. This can be attributed to the complexity of AI projects, which require long-term commitment and continuous availability of employees to achieve effective and efficient results. Furthermore, in an emerging field such AI, where talent is scarce, companies seek to retain the best professionals by offering full-time jobs. This not only ensures continuous dedication to complex AI projects, but also aims to attract and retain highly qualified experts in a competitive market.

As regards industrial sectors, Fig. 2 shows the distribution of the demand for AI-related jobs across various industrial sectors. The highest concentration of demand is in the "IT Services and Consulting'' sector, which represents 43.85 % of the total, with 456 vacancies, followed by the "Software Development'' sector (10.19 %) and companies in the "Technology, Information, and Internet'' sector (5.10 %). Other sectors such as "Human Resources Services'', "Banking'', "Financial Services'', and "Higher Education'' are also significant. At the same time, it is important to note that there is no significant demand for this type of employment in the traditional productive sectors of the Chilean economy, such as mining, agriculture, forestry, and fishing.

The company leading in terms of the number of job postings is the Argentine-based global software development firm BairesDev, with 216 vacancies, followed by Evalueserve (14 vacancies), SoftServe (13 vacancies), Acid Labs (13 vacancies), and Perceptual Consultores LTDA (12 vacancies), which together make this sector stand out (Fig. 3). The companies that published the job offer were not detailed in 5.45 % of the cases.

4.2. Professional profiles required by companies in relation to AI

The main positions required in Chile related to AI include DS, Data Engineer (DE), Software Engineer (SI), Senior Developer (SD), and DA (Fig. 4). The results of the most in-demand positions for working with AI are consistent with previous findings by Chui et al. (2022). A summary of the functions of these positions according to the job profile descriptions defined in ESCO follows.

The DS hold a predominant position due to their ability to identify and analyze high-value data sources, manage large volumes of information, and ensure data consistency and reliability. They use

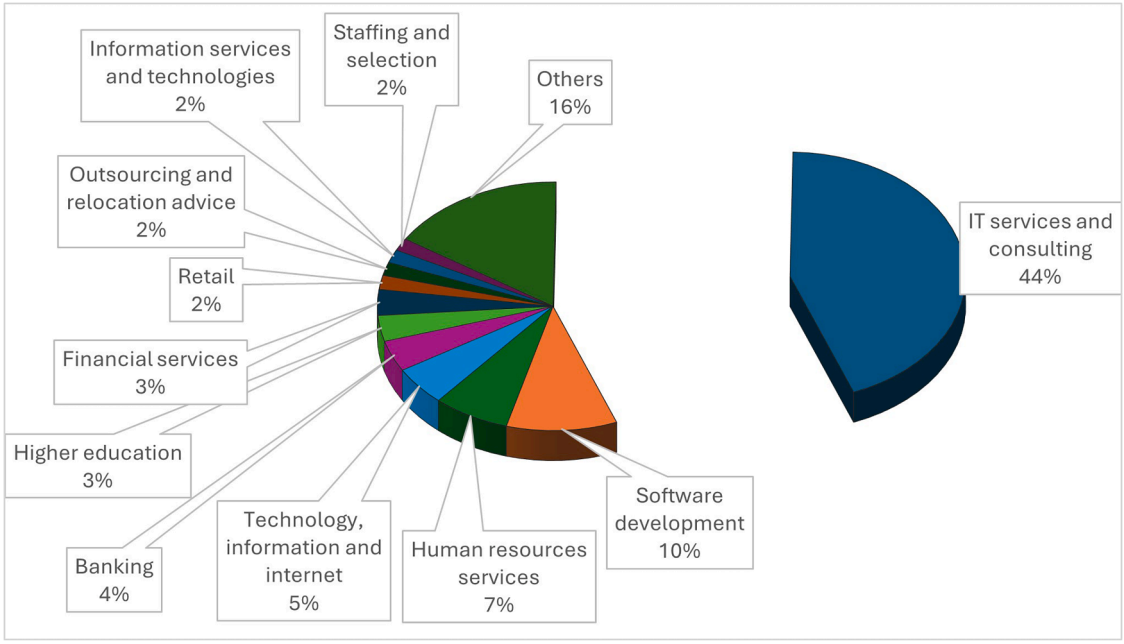

Fig. 2. Sectors with the highest labor demand in jobs related to AI in Chile. **Source:** Own elaboration, 2025.

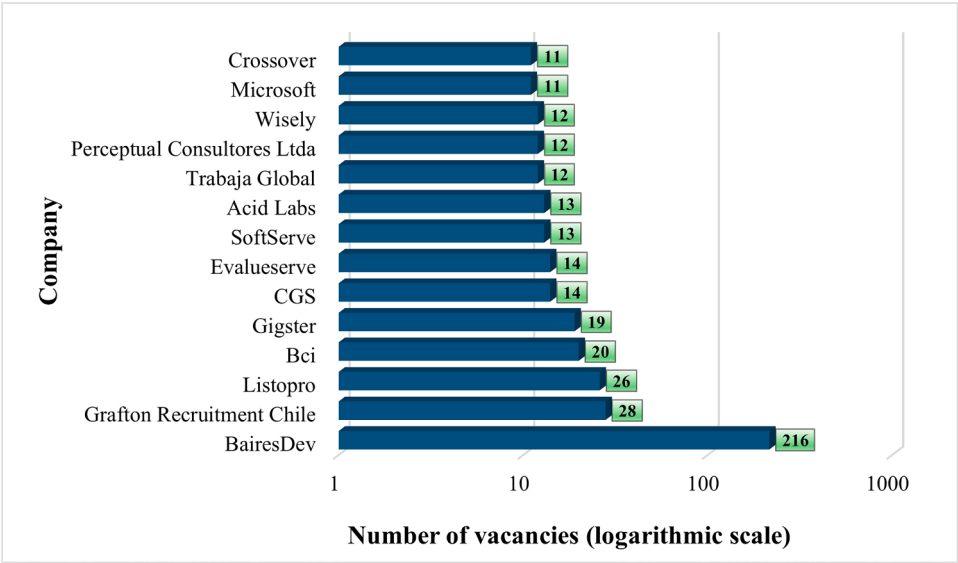

Fig. 3. Companies with the highest labor demand in jobs related to AI in Chile. **Source:** Own elaboration, 2025.

mathematical models to generate visualizations that help to understand underlying patterns and communicate their findings to different audiences, thereby contributing to informed decision-making (European Commission, 2024).

The DE plays a crucial role in data-driven decision-making. They collect, transform, and publish data, design and operate data processing systems, and ensure these systems are secure, efficient, and reliable. Additionally, they implement and train existing ML models (European Commission, 2024). Moreover, this profile requires technical skills such as databases, big data, cloud computing, data science, data warehouse, software development, development platform, among others (Mamani, 2022).

In the case of SI, they research, design, and develop software systems in collaboration with engineering personnel. They evaluate interfaces between hardware and software, oversee software testing and validation, and make modifications to correct errors and enhance performance. They also develop documentation and maintenance procedures, and consult with clients on system maintenance (European Commission, 2024).

The SD implements systems according to specifications and designs using various programming languages and tools (European Commission, 2024). The main difference between "Senior'' and "Junior developers (JD)" lies in the experience and responsibility assumed. "Senior" roles typically require 4.7 years of experience, whereas "Junior" roles require 1.9 years (Cico et al., 2021).

DAs import, clean, validate, model, and interpret datasets to meet business objectives. They ensure coherence and reliability of data sources and generate visual reports such as graphs, tables, and dashboards (European Commission, 2024).

Lastly, occupations such as 'ML engineer' or 'AI engineer,' or 'AI developer' are not specifically listed in the ESCO database. However, they are closely related with the occupation known as 'designers of

Fig. 4. Word cloud for the most in-demand position titles for AI employment in Chile. **Source:** Own elaboration, 2025.

intelligent information and communication technologies systems (DIICTS)' within this database. These professionals apply AI methods in engineering, robotics, and computer science to design programs that simulate intelligence. This includes developing thinking models, cognitive systems, and knowledge-based systems aimed at problem-solving and decision-making. Additionally, DIICTS are responsible for integrating structured knowledge into computer systems, such as ontologies and knowledge bases (European Commission, 2024).

Regarding educational level, the results indicate a clearly defined pattern towards university professionals in 94.2 % of the vacancies, with 4.18 % requiring postgraduate studies, totaling 98.38 %. In contrast, only 1.45 % of the vacancies require high school education, and 0.2 % require technical education. The trend towards demanding potential employees with postgraduate degrees can mainly be seen in multinational companies (IBM, Visa, Amazon Web Services, Uber) and universities seeking lecturers and researchers (Universidad Adolfo Ibáñez, Universidad Diego Portales, and Universidad Católica del Norte). Postgraduate degrees typically required are in computer science or related fields.

With respect to the most sought-after university careers (Fig. 5), Computer Engineering is mentioned in 291 postings and is thus present in 30 % of the total vacancies (961) specifying academic conditions. Secondly, Industrial Engineering is mentioned as a requirement in 13 % of the total set of vacancies analyzed. Additionally, it should be noted that most vacancies do not specify a specific major as exclusive but require at least one related major or specialization.

On the other hand, it is noteworthy to observe that certain educational disciplines, such as Mechatronic Civil Engineering and Process Engineering (which are currently new, rarely recognized careers) are part of this ranking. This phenomenon suggests the possibility that in the future, due to the significant growth projected in the AI job market, these careers may gain greater prestige (Fig. 5). Additionally, it is important to note that 60.27 % of job postings do not specify the university degrees required for these positions, which suggests that there is

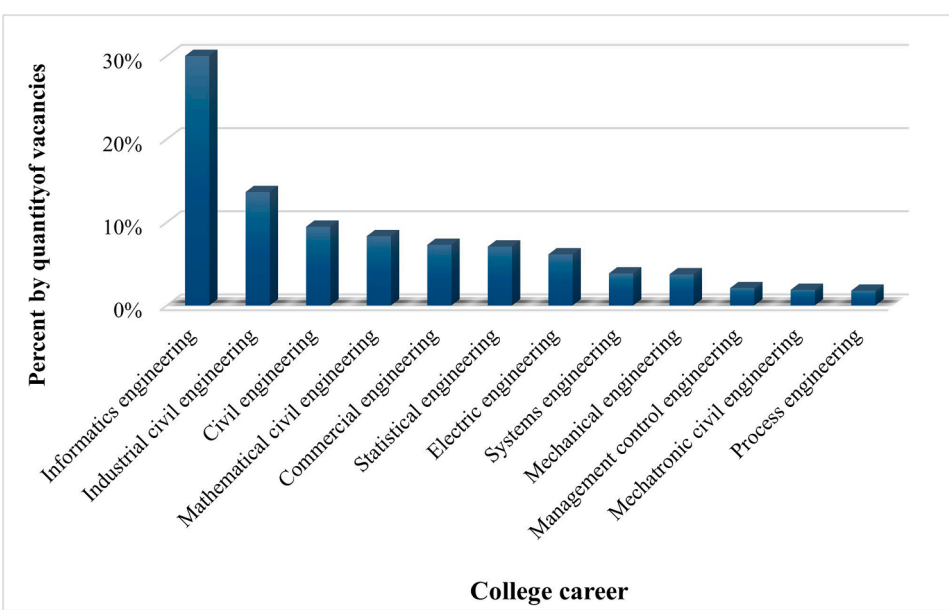

Fig. 5. Most in-demand careers in jobs related to AI in Chile. **Source:** Own elaboration, 2025.

a certain degree of flexibility in educational requirements and a growing emphasis on skills and competencies over formal academic credentials for this type of employment.

As for the required experience, results suggest that it does not seem to be a very decisive factor in all cases. The analysis of the measures of central tendency highlights that the distribution mode is 3 years, indicating that most of the job offers require, at least, this level of work experience. This value represents 24.27 % of the positions examined. Considering the lower and upper limits reveals that 6 % of vacancies require no work experience at all (lower limit: 0 years), while only 0.1 % require 12 years of experience (upper limit). In addition, required experience was not specified in 0.9 % of the postings, further suggesting relatively flexible experience requirements.

Going into the specific skills, Table 2, reveals a set of 15 AI-related skills that are highly demanded in Chile. It is important to note that most of these skills are associated with an ESCO classification code (which will be included in parentheses after each skill in order to facilitate the presentation of the results).

Firstly, the two most transversal skills correspond to the soft-skills category and with four of these skills they are among the 15 most in-demand skills. The soft-skill 'working in teams (S1.8.1)', is a requirement in 617 online job postings, representing 56.1 % of the job positions in the sample. The second most in-demand soft-skill is English 'language proficiency (T1.1)', followed by 'communicating (T4.1)' and 'problem solving (S1.9)' in sixth and seventh position, respectively. It is important to note that proficiency in English is a prerequisite for many jobs in various professions (Tushar & Sooraksa, 2023). This is even more relevant in the advanced context of AI. Therefore, career progression and the use of AI-related software is closely linked to the acquisition of proficiency in this language. Other languages such as Italian, Greek, Portuguese and Mandarin Chinese (T1.1) are also requested, although to a lesser extent.

However, hard skills are generally in the highest demand. A comprehensive analysis of all the data reveals that most skills fall into the category of "technical skills" (71.25 %) as opposed to "soft skills" (28.75 %). Within the set of hard-skills, it is observed that the 'Python programming language (K)' is the third most required skill, with 584 online job offers mentioning it, which is equivalent to 53.1 % of the job offers in the sample. This is followed by the skills 'manage databases (S2.3.0)', 'agile methodology (K)' and 'lean methodology (K)', which occupy the fourth, fifth and eighth positions, respectively.

The ninth to the fifteenth positions are only skills classified as hard-skills. SQL (K) and SQL Server (K) stand out in this category, the former being a generic SQL query skill, although it is distinguished from the latter by the specific language variant. This distinction is based on the need to accurately represent the data collected; therefore, when the SQL query language is mentioned without specifying the variant (MySQL, SQL Server, among others), it is classified as generic.

Table 2
Most demanded skills in AI-related jobs in Chile.

Ability	Category	Quantity	%
Working in teams	Soft	617	56.09 %
English	Soft	611	55.55 %
Python	Hard	584	53.09 %
Database management	Hard	441	40.09 %
Agile project management	Hard	370	33.64 %
Communicate	Soft	368	33.45 %
Troubleshooting	Soft	335	30.45 %
Lean project management	Hard	298	27.09 %
Software and application development and analysis	Hard	278	25.27 %
SQL	Hard	239	21.73 %
Big Data Analysis	Hard	196	17.82 %
Statistics	Hard	188	17.09 %
SQL Server	Hard	178	16.18 %
JavaScript	Hard	176	16.00 %

Source: Own elaboration, 2025.

5. Discussion

The discussion of this paper focuses on two fundamental issues: first, the analysis of the SBTC and RBTC hypothesis based on the results of the present study and second, the relevant characteristics of AI-related employment demand in Chile and other social, geographical and policy outcomes, including the Chilean AI-linked training plan strategy.

5.1. Contribution to the debate on the SBTC and RBTC hypothesis

A core issue for workers with AI in the workplace is the loss of jobs (i.e. the 'Robo- Apocalypse from job loss') (Zirar et al., 2023). Over the last four decades between 50 % and 70 % of changes in the United States wage structure are accounted for by relative wage declines of worker groups specialized in routine tasks in industries experiencing rapid automation (Acemoglu & Restrepo, 2022). AI can increase inequalities in the labor market by augmenting skills used in high-skill jobs, while having relatively more adverse impacts on other jobs. These impacts are expected to be both destructive and transformative (Fossen & Sorgner, 2019; Holm & Lorenz, 2022). Frontline jobs are expected to increase the largest growth (absolute terms). Technology-related jobs are the fastest growing (in %), including Big Data specialist, Fintech Engineers, AI and Machine Learning Specialists and Software and Application Developers. Instead, Clerical and Secretarial Workers (e.g. Cashiers and Ticket Clerks, Administrative Assistants and Executive Secretaries) are expected to see the largest decline (absolute numbers). Similarly, businesses expect the fastest-declining roles to include Postal Service Clerks, Bank Tellers and Data Entry Clerks (Leopold et al., 2025). It is also detected a broad-based increases in employment in high-skill and low skill occupations relative to middle-skilled occupations (i.e., "job polarization") (Acemoglu & Autor, 2011).

Part of the debate on the computerization and automation of economies has focused on aspects such as worker replacement and increasing wage gaps (Holm & Lorenz, 2022; Sohag et al., 2024; Wang et al., 2021). Our results indicate that AI-related job demands are oriented towards professionals with high technical IT skills and university studies that allow them to work in highly complex technological environments. This pattern is consistent with the SBTC hypothesis (i.e. that technological advances increase the demand for highly skilled workers) (Holm & Lorenz, 2022). In the USA, it has been evidenced that AI-related skills can lead to wage premiums of up to 11 % (Alekseeva et al., 2021). This raises the question of whether there has also been an increase in the salaries of this type of professional. Considering that data on wage bands in Chile were not available, this fact should be analyzed in greater depth.

Similarly, the main companies demanding AI-related employment (Fig. 3) and the sectors to which they belong (Fig. 2), are fundamentally linked to software development, information and internet, and banking and financial services. Within these areas, they are mainly focused on automating customer service processes, administrative activities and management, which may be related to routine activities (Martins-Neto et al., 2024). In other words, there is no direct participation of labor-intensive companies related to manufacturing, transportation and activities that require physical and dynamic efforts, nor of key sectors of

the Chilean economy, such as mining, agriculture, construction, and fishing. However, it is possible that companies in these sectors receive these AI services through the hiring of companies specialized in IT services and consulting, since the latter lead in the publication of vacancies. If this was the case, the substitution of workers due to the automation of processes through the incorporation of AI in Chile could be accentuated for administrative and management jobs, such as, customer service, tele-sales, telemarketing, billing, collection, insurance, planning, traceability, document management, etc., which seems to be in line with the proposition that workers in middle-skill activities are negatively impacted by the technological transition, especially those working in routine activities as stated by the RBTC hypothesis (Acemoglu & Autor, 2011; Martins-Neto et al., 2024).

On the other hand, the impact on low-skilled workers in the manufacturing and logistics sectors is difficult to estimate. However, some researchers indicate that AI has improved the accuracy of the supply chain, improving demand forecasting, avoiding disruptions and optimizing inventory levels. At the same time AI-driven automation in these processes has simplified routine tasks, such as order processing, inventory replenishment, and route optimization (Oluwademilade et al., 2024). It may be expected that some logistics workers could benefit from the synergy of efficient administrative and management processes in increasing productivity, which may lead to a higher level of activity, which may also lead to a growing number of jobs. An example of this are the workers of applications that demand this type of employment, such as Amazon Web Services and Uber (the latter is in high demand in Chile). In any case, the full effect of this tendency on the salaries and working conditions of this type of job is yet to be further analyzed

In general, AI-related jobs in Chile seems to be aligned with the pattern stated by Acemoglu and Autor (2011) that posits an increase in employment in high-skill and low-skill occupations compared to medium occupations, especially when they are complementary (Acemoglu & Autor, 2011). However, in line with Albinowski & Lewandowski (for Europe), we also argue that future advances in AI subfields such as robotics could also impact the Chilean labor market with low-skilled workers and routine work being replaced (Albinowski & Lewandowski, 2024), which may cause a change in the structure of skills, fundamentally in industries where the AI-driven transformations are more intensive (Hu et al., 2024). In brief, our findings suggest that the cited "job polarization" may also be currently occurring in the Chilean workplace, however, further studies are required to confirm these trends.

5.2. Social, educational, geographic and policy outcomes of AI in the workplace and the Chilean AI-linked training plan strategy

Given the size of the Chilean labor market, and that the economy only grew by 0.2 % in 2023 and the unemployment rate reached 8.7 % in November of the same year (increasing 0.8 percentage points in the last twelve months) (INE, 2024), results of the study reveal a dynamic and strategic situation. It can be argued that the 1,100 vacancies advertised are a significant result, especially if we compare this number of vacancies with those collected by Verma et al. (2022), that came to 1,200 ads for AI and 1,700 for ML between July and December 2019, for a labor market as large as that of the USA. This indicates that in Chile, demand for AI-related jobs is beginning to occupy a significant place, which is consistent with the upward trend in the number of positions requiring AI skills since 2010 identified by Alekseeva et al. (2021).

In line with other studies, our results show the growing need for AI-related jobs. Even though these type of job vacancies are gaining in importance in different countries such as in the USA (Alekseeva et al., 2021; Verma et al., 2019, 2022), United Kingdom (Samek et al., 2021), Peru (Mamani, 2022), Italy (Colombo et al., 2019), etc., at the same time, they tend to be concentrated in certain states, provinces or municipalities within each country. For example, in the United Kingdom, they are concentrated in London, while in the USA, they are mainly located in California and also in the metropolitan areas of Seattle, Boston and New York (Samek et al., 2021; Verma et al., 2022). This is also the case in Chile, where most of the vacancies are concentrated in a few localities, i.e more than the 56.8 % of job vacancies are concentrated in the capital. This territorial pattern of AI vacancies is related to the sectoral distribution of companies and the productive matrices of Chile's regions. In other words, the sectors that concentrate on demand are closely related to knowledge-intensive activities, services, finance, and banking (Fig. 2), which are mainly located in the capital because it has better digital infrastructure, greater connectivity, and a high concentration of corporate headquarters. Conversely, traditional productive sectors such as mining, agriculture, and fishing, which play a marginal role in the observed demand for AI employment, are the fundamental pillar of productive activity for other regions.

Skills are key drivers of employability and wages, over and above formal education (OECD, 2024b). The expected skill disruption calls for continuous reskilling and upskilling of workers to avoid a collective failure to adjust to skills (Zirar et al., 2023). As explained by McDuie-Ra & Gulson (2020), AI is mainly developed in the Global North and in the technology hubs of East Asia, while the rest of the world is only marginally involved in its development. However, its disruptions and impacts will inevitably occur worldwide, as fragments of labor contributing to its design and programming are globally distributed (McDuie-Ra & Gulson, 2020). In the last decade, many countries have responded to the skill challenge by developing different National Skills Strategies. These strategies must be based on close collaboration and coordination across government and key stakeholders, as well as on a long-term and holistic perspective on skills policies (OECD, 2024a), where companies should also be involved in identifying AI skills and training workers to meet the skill challenge (Morandini et al., 2023). According to the OCDE Skill strategy, the most effective interventions designed for low/medium-skilled workers is demand-driven in coordination with companies, including apprenticeships, dual training and sectoral agreement (OECD, 2019, 2024a). Ahmad (2020) proposes future scenarios of education that involve collaborative styles (e.g. human-machine cooperation, smart virtual active learning campuses and living knowledge learning environments) which may produce desirable benefits for education stakeholders in the future. Meanwhile, automation, AI and the advent of 5G network technologies will drive customization and personalization in higher education delivery and revolutionize the labor landscape in the immediate future (Ahmad, 2020).

Zirar et al. (2023) identify important gaps in the literature concerning training strategies to support skill adjustment and unlearning processes, the retention of highly skilled workers, and the long-term sustainability of "high-skilled" status as technologies evolve (Zirar et al., 2023). Beyond these challenges, a growing body of evidence highlights the importance of territorial context in shaping the labor market impacts of AI, suggesting that policy responses cannot be uniform across regions. Previous research shows that AI adoption and its employment effects vary significantly according to infrastructure endowments, production structures, and industry composition (Clifton et al., 2020; Felten et al., 2021), as illustrated by empirical evidence from the United States (Acemoglu & Restrepo, 2020; Muro & Liu, 2021) and global assessments such as the AI Index 2025 Annual Report (Maslej et al., 2025).

In this regard, a fundamental aspect regarding the challenges faced by the Chilean AI training plan concerns the determination of the required skills for this sector. The competitiveness of trained professionals will depend on the different programs' ability to identify the skills that are most in demand by employers considering Chilean local context. Currently, the main hard-skills (Table 2) demanded in Chile, largely coincide with those that were identified in other research, highlighting: Python, Java, SQL and C++ programming languages (Samek et al., 2021). Verma et al. (2022) make distinctions indicating that ML jobs require more hard skills of software programming and

statistical knowledge, data analysis, and ML skills, such as outlier detection, regression, classification, etc. In contrast, AI-related jobs assign more importance to the soft skills of communication, teamwork and senior management (Verma et al., 2022). Although soft skills are less representative in number in the case of Chile, it can be observed that there are some that are transversal, as is the case of the ability to 'teamwork' and 'English language proficiency', while other studies also indicate that other soft skills (such as communication skills, problem solving, creativity and writing), are highly valued (Gardiner et al., 2018; Samek et al., 2021). However, at least in the Chilean case, it seems that soft skills are highly valued mainly as a complement to hard skills.

Although recent literature highlights efforts aimed at integrating components related to empathy, ethics, and emotional intelligence into AI algorithms, seeking greater similarity to human behavior (Alanazi et al., 2023; Subramani & Manoharan, 2024), human ability to generate unique and divergent responses to seemingly similar situations constitutes a distinctive feature of human evolution usually driven by emotions (Zhou, 2021). Particularly relevant interaction of AI occurs in those occupations where empathy, ethical judgment, creativity, emotional intelligence, and Social Skills are essential to achieving optimal performance (Canbul, 2025; Luckin, 2024). Therefore, occupations involving genuine human interaction, such as education, healthcare, psychology, law, talent management, leadership, and social support, where empathy is fundamental, can hardly be replaced under the technical principles of AI (Montemayor et al., 2022; Subramani & Manoharan, 2024). On the other hand, AI may become a powerful tool for diagnosis, data analysis, process automation, and monitoring (Olawade et al., 2024), human professionals will remain irreplaceable in building trust, connecting with people's deeper motivations, and ensuring ethical conduct based on regulated and conscious empathy (Subramani & Manoharan, 2024), It therefore seems necessary to strike a balance between the recent increase in technological skills and certain "human-only skills" in order to maximize the potential of new employees in the field of AI.

Finally, beyond the debate on whether AI will effectively integrate these "human-only skills" (Chowdhury et al., 2023; Jaiswal et al., 2022; Leopold et al., 2025; Zirar et al., 2023), it is important to consider that the replacement of such jobs will face legal restrictions (Olawade et al., 2024), social scrutiny, and resistance from unions and professional associations. Consequently, these occupations are unlikely to disappear in the short or medium term; rather, they are expected to evolve toward the principles of Industry 5.0, characterized by complementarity between humans and machines (Olawade et al., 2024; Ruiz-de-la-Torre et al., 2023). Under this new paradigm, the training of these professionals should strengthen the strategic use of AI to enhance their technical skills and improve both productivity, effectiveness, and the satisfaction of the individuals they serve, whether patients, students, clients, or users (Olawade et al., 2024; Subramani & Manoharan, 2024), especially for middle-skill workers, in order to reduce the potential impact of "job polarization" in the country.

6. Conclusions, limitations and future lines of research

This study shows that demand for AI-related jobs in Chile remains concentrated in knowledge-intensive service sectors (particularly IT and consulting), largely led by foreign firms with a global presence, while traditional productive sectors such as mining, manufacturing, agriculture, and construction remain only marginally involved. This pattern suggests that Chile is currently at an early stage of AI diffusion, driven mainly by digital service providers and intermediaries rather than by its core productive industries.

The demand for AI-related jobs shows a significant proportion of specialized professional profiles, with university degrees and/or postgraduate qualifications. However, the data points to a competency-based recruitment model that prioritizes high technical skills, applied knowledge, and certain social skills, without ignoring formal credentials. This finding is consistent with the SBTC hypothesis, while the growing emphasis on automation-related roles may, in the medium term, reduce demand for routine mid-skill administrative occupations, in line with the RBTC framework. Nevertheless, further empirical research is needed to assess the magnitude and distributional effects of these dynamics in the Chilean labor market.

From a territorial perspective, AI-related jobs demand is highly geographically concentrated in the capital region, reflecting persistent patterns of spatial centralization. While the available data does not allow verification of actual hiring outcomes, this concentration may exacerbate regional labor market inequalities if not addressed through territorially balanced training and development policies. Differences in regional production structures, infrastructure availability, and innovation ecosystems are likely to shape both the intensity and the nature of AI's labor market impacts across the country.

These findings highlight the need for coordinated policy responses involving Chile's National Training and Employment Service (SENCE, for it acronym in Spanish), educational institutions, and firms. In particular, short-term, modular, and demand-driven training programs (developed in close collaboration with employers) could support rapid upskilling and reskilling, integrate technical and transversal skills, and account for regional disparities and the opportunities offered by remote work. The skill structure identified in this study provides a useful baseline, while data-driven approaches such as the one proposed here are essential to ensure that training systems remain aligned with the evolving AI labor market.

Finally, this study is subject to limitations inherent in the use of online job posting data, including platform heterogeneity, incomplete standardization, and the absence of information on wages, gender, and inclusion. Moreover, job postings capture only part of AI-related jobs demand and do not reflect internal retraining, outsourcing strategies, or actual hiring decisions. Future research should therefore incorporate alternative data sources, such as surveys and firm-level interviews, and explore the broader social, productivity, and distributional impacts of AI adoption across sectors, regions, and demographic groups.

Data availability statement

The database of this manuscript is contained in ADDI a publicly available repository. ADDI is the digital research archive of the University of the Basque Country (UPV/EHU). The data file is available at: http://hdl.handle.net/10810/69075.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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